

A Review of the Applications of CFRP Reinforcements in Civil Engineering

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Abstract. Carbon fiber reinforced polymer (CFRP), as an advanced composite material, has been attracting a great deal of attention from researchers because of its excellent mechanical properties and durability. CFRP reinforcements are increasingly used in the construction and strengthening engineering. In recent years, considering the application of CFRP reinforcements in civil engineering has entered a stage of rapid development, it is necessary to review the development and recent advances in civil engineering. The present paper reviews the applications of CFRP reinforcements in civil engineering, the characteristics and types of CFRP reinforcements and cables, as well as the anchoring system of CFRP cables, are summarized. Furthermore, the applications of CFRP reinforcements and cables in the construction and strengthening engineering are reviewed, especially the developments and applications in bridge engineering. The main goal of this research is to provide a comprehensive review of the application development of CFRP reinforcements in engineering structures. Moreover, the critical issues should be solved for expanding the application scope of them are discussed.

1 Introduction

CFRP is composed of carbon fiber and polymer matrix materials (such as epoxy resin, polyester, nylon resin, etc.), where carbon fiber provides ultra-high strength and high elastic modulus, and the polymer matrix plays a role in bonding, stress transmission, and protecting carbon fiber. CFRP has showed enormous potential in a variety of engineering fields [1].

CFRP was first used in the aerospace field in the 1960s [2]. For the past few years, manufacturing technology has developed rapidly, material technology is constantly advancing, and material costs have also been continuously optimized. These factors have contributed to the gradual expansion of the engineering application scope of CFRP. At present, CFRP has been studied and applied in the fields of automobile manufacturing, ships and submarines, equipment, and construction engineering.

In recent years, with the deepening of research on CFRP and the improvement of production processes, the applications of CFRP cables have gradually increased. CFRP cables can obviously improve the carrying capacity, enhance the durability and seismic performance of structures. Furthermore, compared with steel reinforcements, there are relatively lower carbon emissions of CFRP reinforcements for bridges [3], and also better recycling performance, for instance, the recycled CFRP can be used in the automotive industry [4]. Therefore, CFRP is more environmentally friendly and sustainable, has been extensively applied in engineering.

2 Material properties of CFRP

2.1 Material properties of FRP reinforcements

The commonly used FRP reinforcements include AFRP, GFRP, and CFRP reinforcement. CFRP has the characteristics of ultra-high strength, excellent creep performance, and low relaxation, but the cost is relatively high. The comparison of mechanical properties among these FRP reinforcements and steel reinforcements is shown in Figure 1 [5].

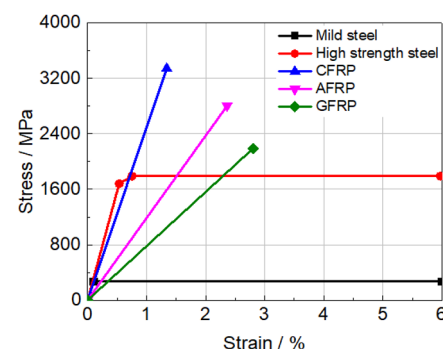


Fig. 1. Strength comparison among FRP reinforcements and steel reinforcements [5].

2.2 Characteristics of CFRP cable

In comparison to traditional cables, CFRP cables have broader application prospects with the characteristics of ultrahigh strength under tensile load, and low weight to

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volume ratio. Table 1 compares several main characteristics of the two kinds of cables [6, 7].

Table 1. Comparison of characteristics between different cables.

Types of cable	Steel cable	CFRP cable
Tensile strength (MPa)	1860	2500
Elastic modulus (GPa)	195	150
Density (g/cm ³)	7.85	1.5~1.8
Maximum longitudinal strain (%)	4.0	1.3~1.6
Expansion coefficient (10 ⁻⁶ /°C)	13.2	-0.9~0.5

The high cost is one of the reasons limiting the widespread application of CFRP reinforcements in engineering structures. However, CFRP reinforcements have superior corrosion resistance, are suitable for complex climatic conditions, and there are few maintenance requirements during the service period of structures [8]. The cost of CFRP reinforcements in the lifecycle and the initial cost of construction can be considered equivalent. However, the cost of traditional materials like steel reinforcements in lifecycle is the sum of the initial cost of construction, the cost of maintenance and repairing of structures in lifecycle, and the indirect cost that caused by the temporary closure when the structures need maintaining, which is always expensive, especially for large-scale projects with the design service life of no less than one hundred years. This makes the application of CFRP reinforcements more advantageous from the perspective of full lifecycle economic performance.

3 The summarize of CFRP reinforcements and cables

3.1 Types of CFRP reinforcements

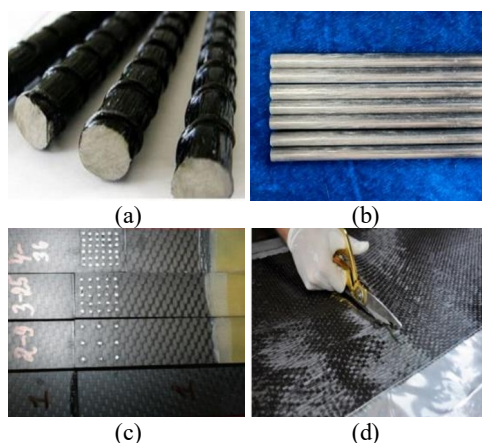


Fig. 2. Different types of CFRP reinforcements: (a) CFRP bars [5]; (b) CFRP rods [9]; (c) CFRP plates [10]; (d) CFRP sheets [11].

See Figure 2, the main types of CFRP reinforcements include CFRP bars [5], CFRP rods [9], CFRP plates [10], and CFRP sheets [11].

3.2 Types of CFRP cables

CFRP cables types mainly include cable of parallel wires [12], cable of twisted wires [13], cable of plates, and cable of nonlaminated straps [14], which are shown in Figure 3. The applications of the above types of CFRP cables in practical engineering will be introduced in the following paragraphs.

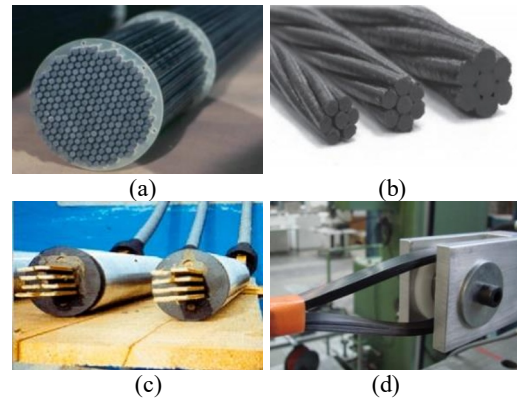


Fig. 3. Types of CFRP Cables: (a) cable of parallel wires [12]; (b) cable of twisted wires [13]; (c) cable of plates [14]; (d) cable of nonlaminated straps [14].

CFRP cables have high axial tensile strength but poor transverse shear resistance due to the anisotropy of the material, the anchoring methods of steel cables are no longer suitable to CFRP cables. Consequently, one of the most important technical challenges of large-scale applications of CFRP cable is that the anchoring system is still very unripe presently, and the anchoring methods will directly affect the utilization effect of CFRP cables, especially for large-span spatial structures. To avoid the damage in the anchoring area and fully utilize the tensile strength of CFRP cable, various anchoring methods have been designed for CFRP cables, which can be generally divided into mechanical anchoring, adhesive anchoring, wedge anchoring, and composite anchoring methods, according to the anchoring mechanism and anchoring performances, see Figure 4 [15, 16].

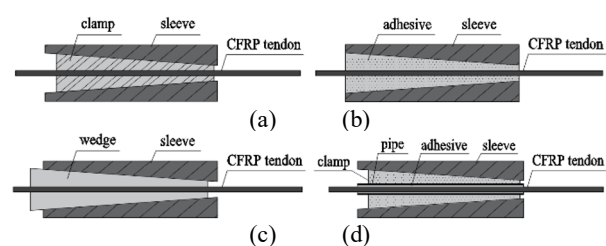


Fig. 4. Types of anchoring methods for CFRP cables: (a) mechanical anchoring; (b) adhesive anchoring; (c) wedge anchoring; (d) composite anchoring method.

4 Research and application of CFRP reinforcement in new construction and strengthening

Because of superior characteristics, CFRP was used as reinforcement materials in CFRP prestressed high-strength concrete components manufactured by

prefabricated manufacturer SCACA, including CFRP prestressed electricity pylons, CFRP prestressed HPSCC (high-performance self-compacting concrete) L-shaped beams, and CFRP prestressed HPSCC columns, respectively [17,18], see Figure 5.

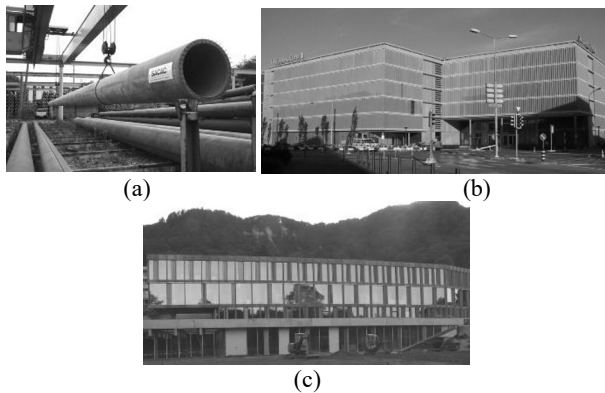


Fig. 5. CFRP reinforcements are applied in construction [25]: (a) CFRP prestressed electricity pylons; (b) CFRP prestressed HPSCC L-shaped beams; (c) CFRP prestressed HPSCC columns.

CFRP reinforcements were also used in strengthening projects. As for the strengthening of buildings, CFRP strips and plates were used to repair the ancient palace which built in 1600 [19], as shown in Figure 6(a), and CFRP rods were used to enhance the aseismic performance of the temple [20], as shown in Figure 6(b).



Fig. 6. CFRP reinforcements are applied in the strengthening of buildings [19,20].

Furthermore, CFRP reinforcements were also applied in strengthening of bridges. A steel truss railway bridge built in Switzerland in 1892, was strengthened using CFRP plates in 2013 [21]. The bridge has a span of 45.2m, a width of 5.0m, and a truss beam height of 6.15m, as shown in Figure 7 (a). There is no need to weld or drill on the existing bridge when using the strengthening system of CFRP reinforcements. Moreover, the system can be disassembled if necessary in future, see Figure 7 (b). The CFRP plate adopted for strengthening work has the elastic modulus of 167.2 GPa and tensile strength of 2710 MPa.



Fig. 7. The steel truss railway bridge strengthened by CFRP plates [21].

The Aabach Bridge was built in 1928 in Lahern, Switzerland, which is a riveted railway steel bridge [22], as shown in Figure 8 (a). In 2020, a new retrofit system using prestressed CFRP rods was used to strengthen the bridge, see Figure 8 (b). The diameter of the CFRP rod is 8mm, the elastic modulus and tensile strength of CFRP plate are 160 GPa and 2047 MPa, respectively. The maximum prestress load applied to the rod is 50kN.

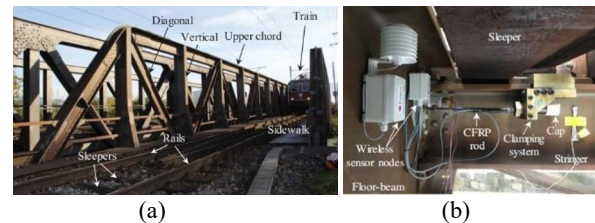


Fig. 8. The steel railway bridge strengthened by CFRP rods [22]: (a) the Aabach bridge; (b) the retrofit system.

5 Applications of CFRP cables in civil engineering

The Komatsu Saito office building is located in Ishikawa prefecture, Japan [20]. In 2016, the building was strengthened using the CFRP rods called CABKOMA, by connecting the roof of the building to the external foundation, which a 160m long CFRP rod weighed just 12kg, as shown in Figure 9. A mesh structure made of CFRP rods was also installed inside for strengthening the building.

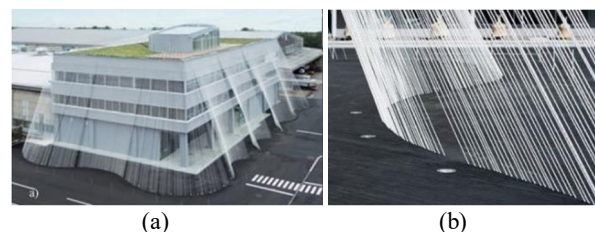


Fig. 9. Office building strengthened with CFRP rods [20].

The Sanya sports stadium in Hainan province was completed in 2021 (see Figure 10) [23]. The stadium adopted a chamfered pentagonal awning of steel structure, and the roof structure consists of a spokewise cable-truss structure and a membrane structure roof, the cantilever length of the roof is 45m. The spokewise cable-truss structure is composed of fully locked ring cables and CFRP plate cables. The sports stadium has become the first large-span spatial structure which applying CFRP cables in the world.

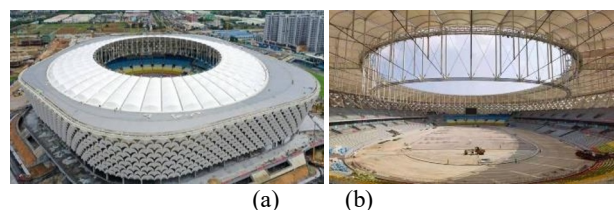


Fig. 10. Sanya sports stadium [23]: (a) the exterior of the stadium; (b) the interior of the stadium.

Up to now, although CFRP cables have been applied to some building structures, they are mainly applied in bridge structures. The following examples can be used to show how CFRP cable applications are developed in bridges.

As the first highway cable-stayed bridge using CFRP cables in the world, Stork bridge in Switzerland was opened to traffic in 1996[12], as shown in Figure 11. The bridge has a A-shaped tower with a height of 38 m, and with span arrangement being (63+61) m. Each CFRP cable consists of 241 5mm diameter wires, and had been cyclically loaded over 10 million cycles in the laboratory, with the load reached three times the allowable load for the bridge.



Fig. 11. Stork bridge using CFRP cables [12].

In 1999, a cable-stayed footbridge was available to the public in the vicinity of the railway station in Herning, Denmark (see Figure 12) [24]. The footbridge was constructed across a railway switchyard, and the spans on either side of the pylon are both 40m. The footbridge was constructed for the purpose of researching and developing the new noncorrodible materials that are suitable for bridge. 16 stays were all made of CFRP cables with the polyethylene coats, protective stainless steel sleeves up to 3.5m above deck were installed to prevent vandalism.



Fig. 12. The cable-stayed footbridge using CFRP cables in Herning [24].

In 2002, The Laroin cable-stayed bridge was constructed in France, as shown in Figure 13 [25], which is the first cable-stayed bridge using CFRP cables in France. It is a cable-stayed pedestrian overpass, with steel main beams and inverted V-shaped steel tube bridge towers, and has a span of 110 m. There are 8 CFRP cables distributed at both sides of the bridge, each cable consists of 7 parallel 6mm diameter CFRP rods.



Fig. 13. Laroin cable-stayed pedestrian bridge using CFRP cables [25].

The cable-stayed bridge with single tower and double cable planes located in Jiangsu University was built in May 2005, which is a footbridge with 16 CFRP stays in total, as shown in Figure 14 [26]. There are three types of stays consisting of six, eleven, and sixteen parallel 8mm diameter CFRP rods, respectively.



Fig. 14. The cable-stayed bridge using CFRP stays in Jiangsu University [26].

The first cable-stayed bridge in Maine, Penobscot Narrows bridge (see Figure 15) was opened to traffic on December 2006 [27]. This bridge was constructed as the emergency replacement for the Waldo–Hancock bridge, which suffered corrosion and premature damage due to the geographical location near the coast. For purpose of assessing and evaluating the application effect of CFRP cables in marine environments, six epoxy coated steel strands were replaced with CFRP cables by the Maine Department of Transportation in June 2007, in addition, these steel cables and CFRP cables were monitored.



Fig. 15. Penobscot Narrows bridge [27].

The three span (72+72+72m) stressed ribbon pedestrian bridge crossing over Júcar river in Cuenca, Spain, was constructed in 2011 to improve the transportation condition of the campus, as shown in Figure 16 (a) [28]. Considering the superior mechanical and durability properties, CFRP cables, see Figure 16 (b), were used in place of conventional cables. The superstructure consists of CFRP cables and precast concrete beam segments, the thickness of bridge deck is 0.34m, and the slenderness ratio of the deck is 211. The bridge is the first application of CFRP cables in stressed ribbon pedestrian bridge.



Fig. 16. The stressed ribbon pedestrian bridge using CFRP cables [28]: (a) the stressed ribbon pedestrian bridge; (b) the CFRP cables.

The public urban transport operator (SSB) in Stuttgart, Germany, constructed a network arch bridge of 124 m

span with CFRP hangers in 2020 (see Figure 17) [29]. It is the first large bridge fully depending on CFRP hangers in the world. The cross-sectional area of CFRP hanger was reduced by 75% compared to what would be required in steel. Moreover, compared with the design with steel hangers, the excellent fatigue resistance makes CFRP hangers a more economical solution.



Fig. 17. The network arch bridge using CFRP hangers in Stuttgart [29].

In 2023, the through concrete-filled steel tube tied-arch bridge (see Figure 18) named Sanduoxi bridge was constructed in Gaoyou, Jiangsu [30, 31]. The bridge is the first tied-arch highway bridge in China using CFRP hangers, with a span of 92.64m, and the deck width of 12.8m. There are a total of 32 hangers adopted in the bridge, four symmetrically positioned hangers of which are CFRP rods, and the rest of them are steel strand rods.

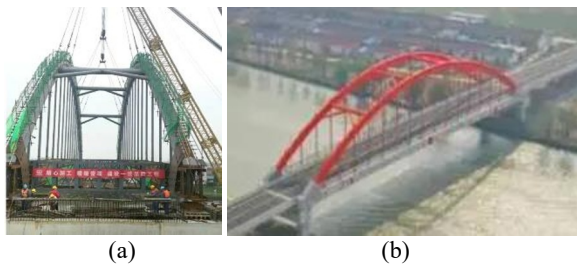


Fig. 18. The tied-arch highway bridge using CFRP hanger rods [30, 31].

In 2023, the cable-stayed bridge with single tower and double cable plane on Huangshan road in Jiangyin, Jiangsu (see Figure 19) using the longest CFRP cable in China [32], was completed with the cable-stayed lifting. The bridge has a span arrangement of (30+100+88+30) m, which is composed of separated steel box girders, herringbone bridge towers, and double saddle shaped cable plane. There are a total of 48 cables on the bridge, forty-six of which are 1770 MPa grade high-strength and zinc-aluminum alloy coated steel cables, the other two are CFRP cables with a length of around 68m and an elastic modulus exceeding 170 GPa.



Fig. 19. The cable-stayed bridge with CFRP cables on Huangshan road in Jiangyin, Jiangsu [32].

The Longmen cross-sea bridge is an important component of the mainline of national highway G228, is also an important highway connecting the Guangxi north bay economic zone [33], which bridge closure was completed in January 2024. The suspension bridge has one span steel box girder of fully floating system, with the main bridge length of 1198m, the main span of 1098m, and the tower height of 174m, as shown in Figure 20, which is the first cross-sea suspension bridge using CFRP cables in the world.



Fig. 20. Guangxi Longmen cross-sea bridge using CFRP cables [33].

The cable-stayed bridge with steel arch tower and double cable plane that cross over Tuhai river in Liaocheng, Shandong. It is the first highway cable-stayed bridge using CFRP cables in China [34, 35], and the span arrangement of main bridge is (100+100) m, see Figure 21. There are two main towers with a height of 55.9 m and one auxiliary tower with a height of 51.9 m, the semi-floating system was adopted in the main bridge. There are 36 pairs of CFRP cables (Φ 7-121) in double cable planes, and are arranged as fan-shaped along the longitudinal direction. The maximum length of these CFRP cables is 73.9 m, and the maximum bearing capacity of the cables exceeds 1000 tons.

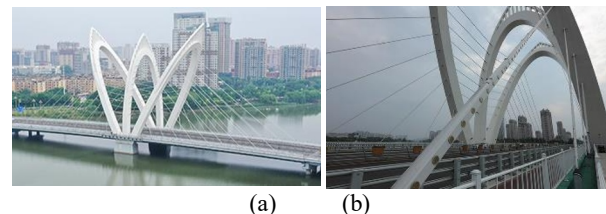


Fig. 21. Tuhai River Bridge using CFRP cables [34, 35].

The Changtai Yangtze river bridge (see Figure 22), with a main span of 1208m, is the largest highway-railway cable-stayed bridge in the world [36]. The bridge is still under construction while the main structure of the bridge located on the Changzhou side had been completed in May 2024. There are twenty-eight CFRP cables (Φ 7-127) arranged on the upstream and downstream, namely seven CFRP cables on each side of the beam, the corbel on deck and crossbeam corbel on the mid span side of the main tower are connected by these CFRP cables.



Fig. 22. Changtai Yangtze river bridge using CFRP cables[36].

6 Conclusion

This article provides a comprehensive review of the application and development of CFRP reinforcements in engineering structures, and draws the following conclusions:

1. The existing research and engineering applications have proved that CFRP reinforcements can not only replace the traditional steel reinforcements but also show better mechanical properties and durability. With the continuous maturity of the manufacturing process of CFRP reinforcements, the production efficiency is improved and the cost could be reduced, and the application of CFRP reinforcements in engineering structures will have a larger scope of development.

2. The main factor currently limiting the widespread application of CFRP reinforcements in structures is the lack of support from normative documents. The application of CFRP reinforcements will be greatly promoted by improving relevant design specifications and regulations and providing corresponding design references.

3. The combination with high-performance materials, including high-performance concrete and high-strength steel, makes it possible to expand the application field of CFRP reinforcements from traditional structural systems to new structural systems with better structural and economic performance.

Acknowledgements

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References

1. Karataş M A, Gökkaya H. A review on machinability of carbon fiber reinforced polymer (CFRP) and glass fiber reinforced polymer (GFRP) composite materials. *Defence Technology*, 2018, 14(4): 318-326.
2. Perner M, Algermissen S, Keimer R, et al. Avoiding defects in manufacturing processes: A review for automated CFRP production. *Robotics and Computer-Integrated Manufacturing*, 2016, 38: 82-92.
3. Liu H Y. Assessment of environmental impact for reinforced concrete bridges and CFRP bridges, *World Bridges*, 2024, 52 (03): 129-130.
4. He D, Kim H C, Sommacal S, et al. Improving mechanical and life cycle environmental performances of recycled CFRP automotive component by fibre architecture preservation, *Composites Part A: Applied Science and Manufacturing*, 2023, 175: 107749.
5. Frhaan W K M, Abu Bakar B H, Hilal N, et al. CFRP for strengthening and repairing reinforced concrete: A review, *Innovative Infrastructure Solutions*, 2021, 6: 1-13.
6. Wang L C. Research on mechanical behaviors of CFRP cables and its connected joint for tension cable roof, Tianjin University, 2017.
7. Zhang P. Study on the mechanical performance and design method of CFRP parallel-rod cables and anchorages, Tsinghua University (2014).
8. Yang Y, Fahmy M F M, Guan S, et al. Properties and applications of FRP cable on long-span cable-supported bridges: A review, *Composites Part B: Engineering*, 2020, 190: 107934.
9. Liu X, Su Q, Zhu J, et al. The aging behavior and life prediction of CFRP rods under a hygrothermal environment, *Polymers*, 2023, 15(11): 2490.
10. Arnautov A K, Kulakov V, Andersons J, et al. Experimental investigation on stiffness and strength of single-lap z-pinned joints in a laminated CFRP stress-ribbon strip, *The Baltic Journal of Road and Bridge Engineering*, 2016, 11(2): 120-126.
11. Hemeda S. Carbon fiber reinforced polymers (CFRP) for strengthening and seismic retrofitting of historic circular masonry stone columns, *Advances and Challenges in Structural Engineering: Proceedings of the 2nd GeoM East International Congress and Exhibition on Sustainable Civil Infrastructures, Egypt 2018–The Official International Congress of the Soil-Structure Interaction Group in Egypt (SSIGE)*. Springer International Publishing, 2019, 114-137.
12. Meier U. Carbon fiber reinforced polymer cables: Why? Why not? What if? *Arabian Journal for Science and Engineering*, 2012, 37: 399-411.
13. Zdanowicz K, Kotynia R, Marx S. Prestressing concrete members with fibre-reinforced polymer reinforcement: State of research, *Structural Concrete*, 2019, 20(3): 872-885.
14. Motavalli M, Czaderski C, Pfyl-Lang K. Prestressed CFRP for strengthening of reinforced concrete structures: Recent developments at Empa, Switzerland, *Journal of Composites for construction*, 2011, 15(2): 194-205.
15. Schmidt J W, Bennitz A, Täljsten B, et al. Mechanical anchorage of FRP tendons—a literature review, *Construction and Building Materials*, 2012, 32: 110-121.
16. Li Y, Shi T, Qiu Y, et al. Anchorage research for CFRP tendons: a review, *Materials*, 2024, 17(13): 3208.
17. Terrasi G P, Brönnimann R, Bättig G. CFRP tendon prestress monitoring by resistance measurement in a spun concrete powerline pylon, *Proceedings of the first middle east conference on smart monitoring, assessment and rehabilitation of civil structures (SMAR 2011)*, American University, Dubai, UAE, 2011, 810.
18. Terrasi G P. Prefabricated thin-walled structural elements made from high performance concrete prestressed with CFRP wires, *Journal of Materials Science Research*, 2013, 2(1): 1.

19. Bastianini F, Corradi M, Borri A, et al. Retrofit and monitoring of an historical building using “Smart” CFRP with embedded fibre optic Brillouin sensors, *Construction and building materials*, 2005, 19(7): 525-535.
20. Ghafoori E, Motavalli M, Nussbaumer A, et al. Design criterion for fatigue strengthening of riveted beams in a 120-year-old railway metallic bridge using pre-stressed CFRP plates, *Composites Part B: Engineering*, 2015, 68: 1-13.
21. Heydarinouri H, Nussbaumer A, Motavalli M, et al. Strengthening of steel connections in a 92-year-old railway bridge using prestressed CFRP rods: Multiaxial fatigue design criterion, *Journal of Bridge Engineering*, 2021, 26(6): 04021023.
22. Franta A, Zajac D. Carbon fiber reinforced polymer and tensegrity structures in search of model architectural and engineering solutions, *Technical Transactions*, 2022, 119(1).
23. Liang C, Zhu Z, Bai G, et al. Design and research on roof structure of the stadium of Sanya International Sports Industry Park, *Proceedings of IASS Annual Symposia. International Association for Shell and Spatial Structures (IASS)*, 2023(4): 1-8.
24. Christoffersen J, Hauge L, Bjerrum J. Footbridge with carbon-fibre-reinforced polymers, Denmark, *Structural engineering international*, 1999, 9(4): 254-256.
25. Dehmous H, Weleman H. Multi-scale reliability analysis of composite structures—Application to the Laroin footbridge, *Engineering Failure Analysis*, 2011, 18(3): 988-998.
26. Mei K, Li Y, Lu Z. Application study on the first cable-stayed bridge with CFRP cables in China, *Journal of Traffic and Transportation Engineering (English Edition)*, 2015, 2(4): 242-248.
27. Jay Rohleder Jr W, Tang B, Doe T A, et al. Carbon fiber-reinforced polymer strand application on cable-stayed bridge, Penobscot Narrows, Maine, *Transportation research record*, 2008, 2050(1): 169-176.
28. Romera L, Hernández S, Baldomir A, et al. Study of pedestrian comfort in a three span stress ribbon footbridge with carbon fibre cables, *WIT Transactions on The Built Environment*, 2020, 196(14): 139-152.
29. Meier U O, Winistörfer A U, Haspel L. World’s first large bridge fully relying on carbon fiber reinforced polymer hangers, *Proceedings of the SAMPE Europe Conference, Amsterdam, The Netherlands*, 2021, 30.
30. Five bridges flying over the river! The waterway regulation project in Gaoyou of Tongyang line has been fully completed by China Railway Tenth Bureau (2023). <https://baijiahao.baidu.com/s?id=1773763638682459669>.
31. Liu L, Zhu Y, Huang S, et al. Design and application of CFRP suspender cable for tied arch bridge, *Modern Transportation and Metallurgical Materials*, 2021, 1 (04): 70-74.
32. The longest carbon fiber composite cable in China made its debut in Jiangyin (2023). <https://www.ccia.xin/guonazixun/1399.html>.
33. The longest cross-sea bridge in Guangxi has fully entered the superstructure construction stage (2023). https://www.gov.cn/yaowen/2023-05/01/content_5753902.htm.
34. Lu C, Wang X, Ning Y, et al. Numerical investigation on the structural behavior of a short-span cable-stayed bridge with steel and CFRP hybrid cables, *Materials*, 2024, 17(9): 2032.
35. Classic Project 2: The bridge crossing over Tuhai river on Xinghua road in Liaocheng | The largest carbon fiber cable-stayed bridge in the world (2024). <https://www.sdcpd.com/news/491.html>.
36. Expected to close in June! The Changzhou side of the Chang-Tai Yangtze river bridge was the first to be connected (2024). https://www.thepaper.cn/newsDetail_forward_27289088.